

Comments on Class Meeting #3

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Some details are important. Please make sure that you understand the mechanistic details presented in these comments. It is important to learn correct habits early in the course.

Answers to Selected Problems

Here I present some comments on the problem set presented in the handout for this meeting. We never have time to do every kind of mechanism in these early review classes. It will be left for you to complete them on your own. I will provide these comments on selected problems because the foundation of organic chemistry is very important.

5. Write the mechanism for base-catalyzed hydrolysis of an ester.

Observe that the hydroxide adds to the carbonyl group of the ester to give the famous tetrahedral intermediate. The intermediate can eliminate the alkoxide to provide the carboxylic acid and the alkoxide. At this moment, the reaction is entirely reversible. However, we will now have rapid proton transfer from the carboxylic acid to the alkoxide to give the irreversible final products, a carboxylate anion and an alcohol. A very weak base and a weak acid are much lower in energy than a weak acid and a strong base.

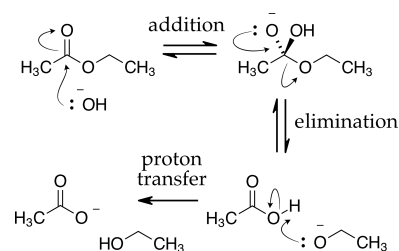


Figure 1: Base catalyzed hydrolysis of an ester.[†]

6. Write the mechanism for acid-catalyzed hydrolysis of an ester.

No strong nucleophile, such as hydroxide, will exist in an acidic water solution. Water is a very poor nucleophile and will not easily add to the carbonyl group. In strongly acidic conditions, we will have a small amount of protonated carbonyl groups in equilibrium. Water can add to this activated carbonyl group. In strong acid, any of the oxygen groups may be protonated, and among these options, the species with the protonated alcohol oxygen will lead to the product. In the end, water can accept a proton from the protonated carbonyl product to give us a carboxylic acid and an alcohol. The reaction is reversible because we regenerated a strong acid. The reverse is the condensation reaction for the formation of an ester. However, equilibrium will favour the products because of the enormous excess of the water reactant, which is the solvent.

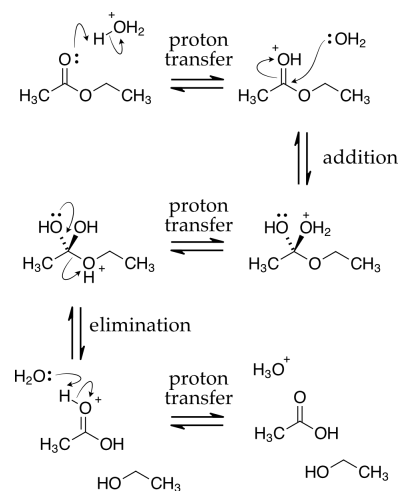


Figure 2: Acid catalyzed hydrolysis of an ester.[†]

7. Now write the mechanism for acid-catalyzed condensation of an alcohol and a carboxylic acid to make an ester. Are they similar? Why is the base-catalyzed option for ester synthesis not a thing?

The reaction is reversible because we regenerated a strong acid. The reverse is the condensation reaction for the formation of an ester. However, the equilibrium will favour the products in water because of the enormous excess of water, which is the solvent.

10.a Write the mechanism for imine formation under basic conditions

Amines in neutral water are likely protonated. The base can deprotonate the ammonium ion to give the neutral amine, a nucleophile. The addition of the carbonyl group gives a zwitterion. This intermediate is high in energy, having a strong base and a weak acid in the same molecule. The formation of this intermediate is a slow step. It quickly relaxes by proton transfer to give a weak acid and a weak base. The alkanolamine product can eliminate a hydroxide, giving the protonated imine a weak acid. Hydroxide is a strong base, and it accepts the proton, producing the final products, an imine and water. This is an example of a condensation reaction.

10.b Write the mechanism for imine formation under acidic conditions

Acidity conditions in water could result in a small amount of protonated carbonyl groups at equilibrium. The protonated carbonyl is an activated electrophile that reacts rapidly with the amine to produce a protonated alkanolamine. It is not a high-energy intermediate as in the base-catalyzed path because we now have a weak base (the alcohol) and a weak acid (the ammonium). This step is fast.

However, most of the amine nucleophile would be in the protonated form in acid. There would be very little neutral nucleophilic amine to act as a nucleophile. There is a "sweet spot" where there is enough acid strength to provide a small amount of the activated carbonyl but not so much that no nucleophilic amine is present.

The protonated alkanolamine can have the extra proton on the amine or on the alcohol. The latter is much less likely as it has a strong acid and a weak base. But enough would exist in the equilibrium so that the protonated alcohol group can be eliminated as neutral water. The α -elimination is a rapid step and gives the protonated imine product. The protonated imine is a thousand-fold more acidic than the protonated amine. Under conditions in which there is a small amount of neutral amine, there will be a significant amount of the imine product.

10.c Compare the two imine formation reactions and consider which is faster. Think about it.

The base-catalyzed route has the amine almost entirely available as a nucleophile. However, the high-energy intermediate product formed by the addition is a slow step. There is a lot of reactant in a slow reaction. The acid-catalyzed route has a very activated carbonyl electrophile and a low energy difference to form the intermediate. However, the amine is much less available in acidic conditions. There is very little reactant in a fast reaction. Is it a tie?

Experimental evidence reveals that imine formation, as shown in the two reaction schemes, is fastest at near pH 5. So the acidic route is more rapid if we don't go too far. The imine reaction is easily reversible and favours the starting materials at equilibrium in dilute water solution. However, in the hydrophobic core of proteins, where water can be excluded, we see stable imines form. You would be blind without them.¹

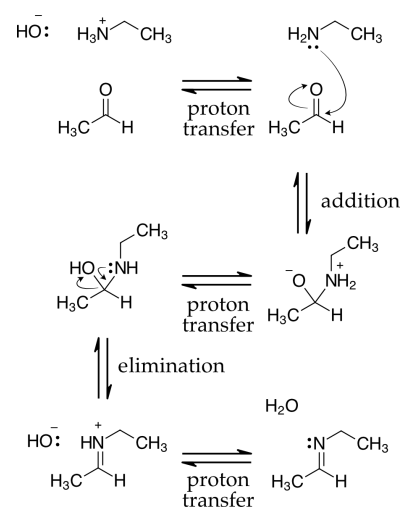


Figure 3: Base catalyzed condensation of an aldehyde and amine to form an imine.[†]

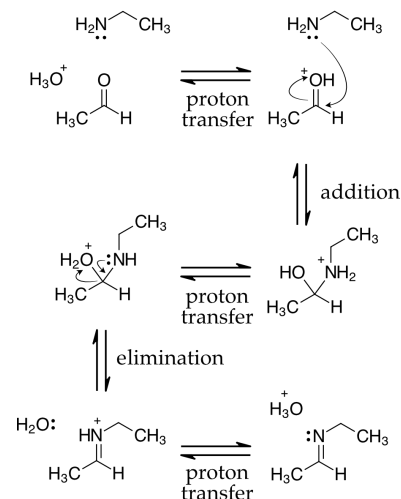


Figure 4: The acid catalyzed version of imine formation.[†]

¹ Rhodopsin requires an imine linkage between the amine of a lysine residue and the aldehyde of the retinal cofactor